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Edge-Enabled Pre-ictal Activity Prediction Framework Using Geometric Deep Learning

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***Abstract*— We present an edge-enabled pre-ictal phase prediction framework for the epileptic seizure geometric deep learning technique. In particular, we utilized the Brain Network Transformer (BNT) for the EEG signals based pre-ictal activity prediction. The model analyzes EEG-based brain connectivity to detect pre-ictal state, offering an interpretable approach compared to traditional deep learning techniques. We present a performance comparative analysis of BNT with various existing techniques. Experimental results prove that BNT outperforms existing techniques. In particular, the BNT was evaluated on the CHB-MIT dataset and achieved an average and median accuracy score of 97.17% and 98.51%, respectively on test data with an average area under the receiver operating curve score of 0.99. Additionally, we benchmark the computational efficiency on an Nvidia Jetson Xavier NX edge device, where it has a mean inference time of 9.978 ms demonstrating suitability for real-time deployment on edge devices. The results highlight the potential of BNT for practical seizure prediction, improving patient outcomes through timely intervention.**

***Clinical relevance*— This study presents an efficient, real-time pre-ictal phase of epileptic seizure prediction system using the Brain Network Transformer, enabling early intervention through EEG-based brain connectivity analysis. With high accuracy and edge-device compatibility, it enhances patient care by facilitating timely seizure prediction, improving quality of life, and supporting practical deployment in clinical and wearable monitoring systems.**

I. INTRODUCTION

Epilepsy is a disorder of the nervous system, defined by the occurrence of recurrent seizures that arise without provocation due to atypical brain activity [1]. It is among the most widespread neurological conditions globally, affecting more than 50 million individuals worldwide. It can affect the patient's quality of life due to its chronic nature and the associated stigmas [2]. One of the major challenges in managing epilepsy is predicting seizures before they occur (pre-ictal state). This early prediction can drastically improve a patient's quality of life by allowing preemptive actions and treatment. The pre-ictal state refers to the period just before a seizure occurs. It is often characterized by subtle changes in brain activity, which may be detectable through advanced monitoring techniques like EEG (electroencephalography). Identifying this state is critical for predicting seizures and implementing timely interventions [3], [4].

Recent advancements in EEG technology such as wearable devices, have made continuous monitoring more feasible. These devices enable long-term tracking of brain activity in real-world settings, which is essential for detecting pre-ictal states [5]. Conventional methods such as visual EEG inspection or simplistic rule-driven systems often lack objectivity, require significant time, and are error-prone. To overcome these limitations, researchers have turned to advanced computational approaches, including machine learning and deep learning algorithms. These techniques can analyze EEG signals and detect pre-ictal patterns. However, they lack explainability and thus, limited clinical potential for prognosis. On the other hand, using a graph-based techniques provides a level of explainability which is necessary for medical applications as they can offer insights into brain activity in terms of connectivity before a seizure event occurs [6], [7].

In this study, we explore the potential of the brain network transformer (BNT) to analyze brain connectivity from EEG data for seizure prediction. We also compare the performance of BNT with previously proposed techniques. Furthermore, we benchmark the computational efficiency of BNT on an Nvidia Jetson platform, assessing its suitability for deployment on resource-constrained edge computing devices.

II. RELATED WORK

Detecting epileptic seizures using EEG signals has been a focus of research for many years. Early studies focused on the analysis of EEG signals using traditional signal processing techniques [8], [9], [10], [11], [12]. While signal analysis based techniques provided valuable insights, they face significant shortcomings. They often required extensive domain expertise for feature engineering. They also struggle with generalizing across diverse datasets due to their reliance on handcrafted features. Additionally, these techniques have been unable to adapt dynamically to the non-stationary nature of EEG signals. These factors have limited the predictive robustness of classical signal analysis based techniques [8]. With the recent advancess in artificial intelligence, deep learning and machine learning algorithms are developed for epilepsy prediction. Early methods combined Recurrent Neural Networks (RNNs) with wavelet decomposition to predict seizures from raw EEG signals [13]. This was followed by the application of machine learning techniques like

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Support Vector Machines (SVMs), L1-regularized logistic regression, and Convolution Neural Networks (CNNs), often combined with novel EGG pre-processing for inter- and pre-ictal classification [14], [15], [16], [17], [18], [19]. Further advancements resulted in deep learning models such as Multi-Layer Perceptron's (MLPs), Deep Convolution Neural Networks (DCNNs), and hybrid CNN-LSTM architectures bypassing preprocessing by using raw EEG signals directly [20], [21]. Generative adversarial networks were also employed to address class imbalance and were paired with CNN and LSTM for classification [22], [23]. A CBAM-3D CNN-LSTM model using Short-Time Fourier Transform (STFT) spectrograms, denoising filters, and attention mechanisms was also developed for accurate seizure prediction [24]. Although traditional deep learning and machine learning techniques achieve low error rates, their "black box" nature limits their applicability in medical settings. This makes them unsuitable for prognosis or understanding the underlying causes of diseases [25], [26]. Geometric Deep Learning (GDL) addresses these shortcomings by leveraging graph-based representations. These representations align with the structure of biological brain data and allow for both accurate predictions as well as deeper analysis of disease mechanisms [27]. In recent times, a lot of progress has been made in developing sophisticated GDL based methods. Multiple studies have explored the potential of Graph Attention Networks (GATs) and Graph Convolutional Networks (GCNs) in combination with innovative graph generation techniques for seizure prediction [28], [29], [30], [31]. In this work, we assess the performance of BNT on scalp EEG data for seizure on-set prediction and explore it's potential applicability in edge computing environments.

III. Methodology

In this study we leveraged the BNT [32], originally designed for imaging data, for epileptic seizure on-set prediction using EEG. BNT is an advanced deep learning architecture designed to model and analyze complex interactions within brain networks. The high level architecture of the proposed algorithm is graphically illustrated in Fig. 1. It utilizes the extraordinary encoding capabilities of transformer models, originally developed for natural language processing (NLP), to encode Brain Networks. We generated brain networks from electroencephalography (EEG) where each electrode in the EEG senor represents a node in the brain network. Edges are established based on the Pearson correlation between the nodes representing the functional connectivity. The BNT functions by generating node-level embeddings for all nodes within the network. These node embeddings are then aggregated using the graph pooling operator called orthonormal clustering readout (OC_{Read}) to produce graph level embeddings. Finally, these graph-level embeddings are then processed by a fully connected neural network (FCN) for classification.

A. Node-level Representation Learning

Unlike other deep learning and Geometric Deep Learning-based methods that leverage a myriad signal processing-based or AI based methods to generate the initial embeddings, we used the connection profile of each embedding of the nodes as their initial embeddings. Connection profiles are the row vectors x from the correlation matrix $X^{N\times N}$. Using the connection profile as the initial embeddings further allowed us to not use positional embeddings as the diagonal element, representing self connection, is always 1. This allows unique node identity in the fully connected graph. However, we set the diagonal of the connectivity matrix to 0 as it helps in focusing the analysis on the actual functional relationships while still allowing unique node identity. The core of BNT's node-level learning lies in its multi-head self-attention layers. These layers dynamically refine node representations by modeling interactions across the entire network. For each layer l and attention head m.

- The input node features Z^0 (initialized as X) are linearly projected into queries, keys, and values using learnable matrices
- Pairwise attention scores are computed for queries and keys.
- The pairwise attention weights are multiplied with the value vectors.

Outputs from all M attention heads are concatenated and linearly projected to generate the final node representations for layer l. This mechanism is represented by (1).

$$\begin{aligned} \boldsymbol{Z}^l &= \left(\mathop{\Big\|}_{m=1}^{M} \boldsymbol{h}^{l,m} \right) \boldsymbol{W}^l_{\mathscr{O}}, \\ \boldsymbol{h}^{l,m} &= \text{Softmax}\left(\frac{\boldsymbol{W}^{l,m}_{\mathscr{Q}} \boldsymbol{Z}^{l-1} (\boldsymbol{W}^{l,m}_{\mathscr{K}} \boldsymbol{Z}^{l-1})^\top}{\sqrt{d^{l,m}_{\mathscr{K}}}} \right) \boldsymbol{W}^{l,m}_{\mathscr{V}} \boldsymbol{Z}^{l-1}, \end{aligned} \tag{1}$$

where $||$ presents concatenation, M is the number of attention heads, $W^{l,m}_Q$,$W^{l,m}_K$,$W^{l,m}_V$ are learnable query/key/value matrices and $\sqrt{d_K^{(l,m)}}$ is the scaling factor.

B. Graph-level Representation Learning

Graph-level embeddings are generated by aggregating the previously generated node-level embeddings using OCR_{ead}. The OCR_{ead} module consists of two stages: (a) orthonormal initialization of cluster centers and (b) soft assignment of nodes to clusters. Firstly, cluster centers are randomly initialized using Xavier uniform initialization [33]. These centers are then ortho-normalized using the Gram-Schmidt process to ensure they are mutually independent. The Gram-Schmidt normalization of the initialized vectors is represented in (2).

$$\boldsymbol{u}_k = \boldsymbol{C}_k - \sum_{j=1}^{k-1} \frac{\langle \boldsymbol{u}_j, \boldsymbol{C}_k \rangle}{\langle \boldsymbol{u}_j, \boldsymbol{u}_j \rangle} \boldsymbol{u}_j, \quad \boldsymbol{E}_k = \frac{\boldsymbol{u}_k}{\|\boldsymbol{u}_k\|}, \tag{2}$$

where C_k represents k^{th} initial cluster with center , u_k is the intermediate orthogonal vector for k^{th} cluster, u_j intermediate orthogonal vector from previous iterations, $\langle \cdot, \cdot \rangle$ presents

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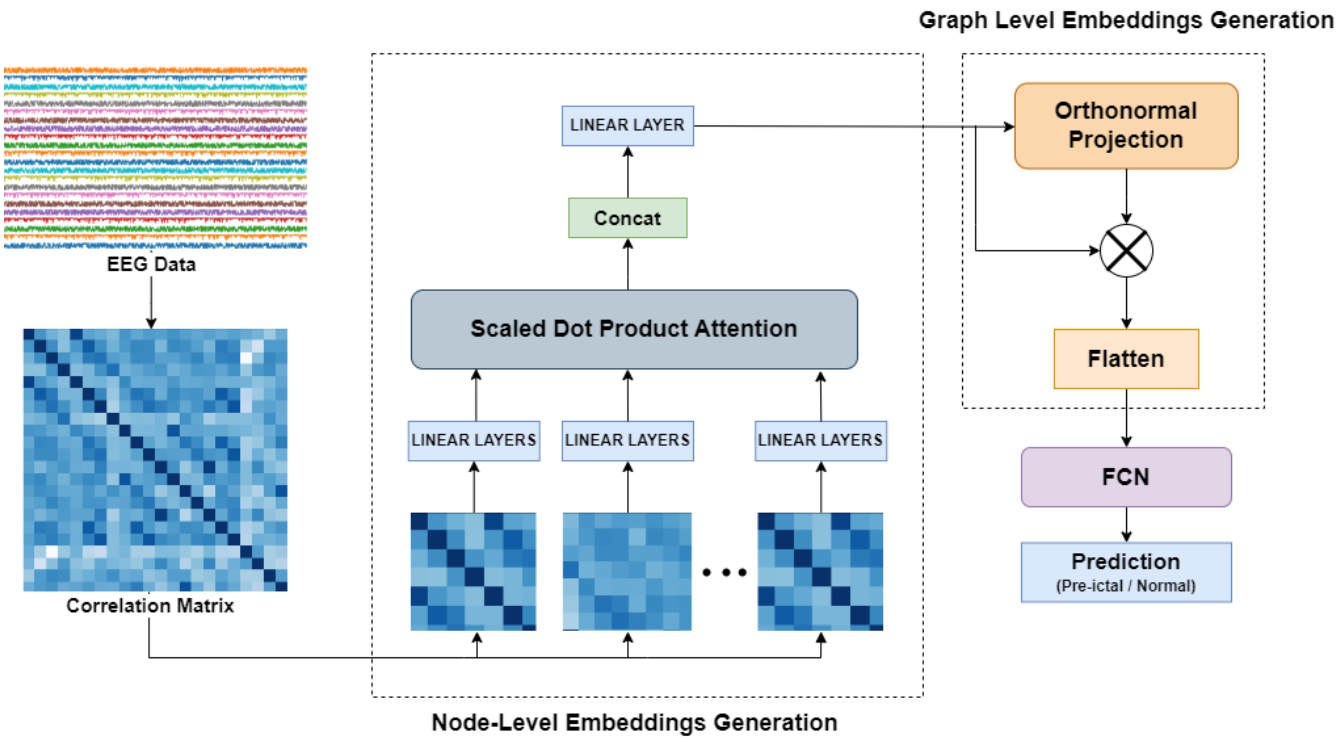

Fig. 1. The architecture of our proposed work predicting pre-ictal phase seizures using EEG signals.

inner product and E_k represents k^{th} orthonormal cluster center.

After the clusters have been initialized, nodes are assigned probabilistically to one or more clusters. This soft assignment is based on the similarity between the node's embedding and the cluster centers. A soft assignment matrix P is calculated based on formula given in (3).

$$\boldsymbol{P}_{ik} = \frac{\exp\left(\langle \boldsymbol{Z}_i^L, \boldsymbol{E}_k \rangle\right)}{\sum_{k'=1}^{K} \exp\left(\langle \boldsymbol{Z}_i^L, \boldsymbol{E}_{k'} \rangle\right)}, \tag{3}$$

where E_k is the kth cluster center, Z_i^L is the ith node of the final MHSA Layer L and P_{ik} the probability of assigning node i to cluster k.

The graph-level embeddings are then obtained by aggregating the node embeddings Z_L with transpose of the soft assignment matrix P. This is expressed in (4). The graph-level embeddings thus generated are passed on to a FCN that is used to classify the pre-ictal state.

$$\boldsymbol{Z}_G = \boldsymbol{P}^\top \boldsymbol{Z}^L, \tag{4}$$

where Z_G are the graph-level embeddings.

IV. Experiments and Results

A. Dataset and Preprocessing

We used the CHB-MIT scalp EEG dataset [37] to assess the potential of BNT for predicting epileptic seizures. This open-source dataset contains EEG recordings from 24 participants affected by epilepsy. The age of these subjects range from 1.5 to 22 years. The dataset contains continuous, long-term EEG recordings and is sampled at 256 Hz. The dataset includes both seizure and non-seizure (interictal) data. The seizure events were annotated by clinical experts. In our study, we only used the recordings with 22 unique channels. Each channel was treated as a node in the brain network. To define the edges between these nodes, we calculated the Pearson correlation coefficients between the time series recordings of each pair of channels. This allowed us to capture the functional connectivity between different regions of the brain. While the ictal data was already annotated, we used the technique employed in [28], [34] to annotate the pre-ictal and inter-ictal period. Specifically, the EEG recordings from 1 hour prior to the ictal period were labeled as the pre-ictal, while recordings from 4 hours before and after the seizure were labeled as inter-ictal. These continuous recordings were then segmented into 10-second samples. We used a 5-second overlap between consecutive pre-ictal samples to deal with the class imbalance.

B. Detailed Performance Evaluation of Brain Network Transformer for Epileptic Seizure Prediction

To evaluate the performance of the proposed technique, we tested it using several metrics namely specificity, sensitivity, accuracy, and the area under the curve (AUC). Table I presents the performance metrics of the proposed approach across individual subjects in dataset. Patients 24 and 12 were left out because they didn't have a pre-ictal state, as defined by [28], [34]. The proposed method performed very well across all metrics, with a mean accuracy of 97.1% and a median accuracy of 98.5%. All patients except 3 achieved an accuracy score of over 96%. Our suggested

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TABLE I

RESULTS OF PERFORMANCE EVALUATION METRICS FOR EACH PATIENT USING OUR PROPOSED BRAIN NETWORK TRANSFORMER BASED METHOD.

Model for Patient	Performance Evaluation Metrics						
	Test Accuracy	*Train Accuracy*	*Test Loss*	*Train Loss*	*Test Sensitivity*	*Test Specificity*	*Test AUC*
chb01	99.7549	99.9540	0.2344	0.0447	0.9958	0.9983	0.9999
chb02	99.2417	99.9287	0.7194	0.0703	0.9748	0.9947	0.9996
chb03	97.2842	97.9815	1.8387	1.5232	0.9695	0.9733	0.9947
chb04	98.3107	98.5580	1.5677	1.1798	0.9306	0.9911	0.9966
chb05	95.3596	97.0058	3.8349	2.6147	0.9441	0.9603	0.9914
chb06	89.5225	89.8393	8.1949	7.5093	0.8897	0.9000	0.9605
chb07	99.1296	99.2833	0.6776	0.6826	0.9665	0.9943	0.9992
chb08	96.5015	98.2353	3.2873	1.4860	0.9805	0.8333	0.9847
chb09	98.7519	99.2062	1.2541	0.6991	0.9733	0.9904	0.9880
chb10	94.6133	96.4606	4.6280	2.9546	0.9572	0.9387	0.9881
chb11	99.4611	99.8167	0.6307	0.1599	0.9864	0.9957	0.9994
chb13	99.1803	100.0000	0.4887	0.0710	0.9800	1.0000	1.0000
chb14	89.1641	91.4062	8.3775	6.4952	0.8951	0.8889	0.9587
chb15	99.2509	99.8106	1.3706	0.1602	0.9957	0.9722	0.9972
chb16	97.5610	98.0634	2.9692	1.9633	0.9583	0.9814	0.9919
chb17	97.9167	99.3924	2.5584	0.5301	0.9722	0.9815	0.9957
chb18	99.8930	99.8793	0.1741	0.0939	1.0000	0.9988	0.9999
chb19	98.7254	98.9971	1.0076	0.8011	0.9014	0.9949	0.9982
chb20	99.3252	99.8311	0.7756	0.1146	0.9600	0.9984	0.9997
chb21	96.2180	97.6144	3.1076	2.0342	0.8310	0.9734	0.9812
chb22	94.6970	95.6913	4.6242	3.1732	0.9175	0.9536	0.9810
chb23	98.8764	99.3845	1.9476	0.4814	0.9726	0.9948	0.9973
Average	97.1696	98.3914	1.4258	0.0700	0.9639	0.9661	0.9978
Median	98.51	99.10	1.7032	0.07500	0.9679	0.9859	0.9969

algorithm also had high Sensitivity, Specificity, and AUC scores, with averages of 0.9639, 0.9661, and 0.9978 across the 22 subjects. This shows that the model works well even with imbalanced datasets, as was the case with numerous subjects. These results are encouraging, especially since no major signal processing steps were involved.

C. Comparative Analysis of Brain Network Transformer with Existing Techniques for Epileptic Seizure Prediction

A comparative analysis was conducted between our method and other techniques as illustrated in Table II. Experimental results demonstrate that the proposed technique outperforms numerous existing methods in pre-ictal state recognition. For example, we achieves a test accuracy of 97.1696%, which is higher than the 91.14% accuracy of the Multitask CNN with Transfer Learning [34]. In terms of AUC, we achieve the score of 0.9908, outperforming methods like Wavelet Transform with CNN [18] (0.8850), CNN [21] (0.9880), and Model Agnostic Meta Learning [23] (0.9680). Our proposed approach also shows higher sensitivity (0.9639) compared to [17] (0.9223), CNN-LSTM [22] (0.9300), and Multitask CNN with Transfer Learning [34] (0.9245). Similarly, its specificity (0.9661) is better than the techniques proposed in [17] (0.9338), [22] (0.9250), and [34] (0.8994). While methods proposed in [35] and [36] also show good sensitivity (0.9651 and 0.9550, respectively), they fall short in other metrics. Overall, our suggested method consistently outperforms other techniques across all key measures, making it a strong model for seizure prediction.

D. Runtime Inference on Edge Device

The computational efficiency of our proposed framework was assessed through runtime analysis on the Nvidia Xavier NX. This would allow us to assess its feasibility for real-time seizure prediction. To ensure reliable results, we repeated the inference process five times and obtained a mean inference time of 9.978 ms. For comparison, we also measured inference speed on a GPU-enabled system with an RTX 4070 Ti, where the model achieved a mean inference time of 5.728 ms. The low inference time on both devices can be attributed to the model's efficient architecture. It eliminates the need for extensive preprocessing and avoids the complexity of multiple deep learning or geometric deep learning models often used in other approaches. These results highlight the potential of deploying the BNT-based algorithm on edge devices for real-time seizure prediction with minimal latency. Fig. 2 compares the average inference time on the Jetson Xavier NX with the GPU-enabled system.

V. DISCUSSION

This study validates the effectiveness of our proposed method in recognizing the pre-ictal state based on scalp EEG data. In particular, BNT's ability to achieve high accuracy (97.1696%) and AUC (0.9908) without extensive signal preprocessing highlights its robustness and generalizability. Furthermore, the BNT's high sensitivity (0.9639) and specificity (0.9661) indicate its ability to handle imbalanced datasets effectively. These results are particularly important given the non-stationary nature of EEG signals and the class imbalance common in seizure prediction datasets. Unlike traditional deep learning models, which are often

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TABLE II

PERFORMANCE COMPARATIVE ANALYSIS OF OUR PROPOSED METHOD WITH EXISTING TECHNIQUES.

Techniques	Test Accuracy	Test AUC	Test Sensitivity	Test Specificity
EMD+SVM [17]	-	-	0.9223	0.9338
Wavelet Transform with CNN [18]	-	0.8850	-	-
CNN [21]	-	0.9880	0.9880	-
CNN-LSTM [22]	-	-	0.9300	0.9250
Multitask CNN with Transfer Learning [34]	91.14	0.9694	0.9245	0.8994
Model Agnostic Meta Learning [23]	-	-	0.9680	0.9550
GCN [35]	-	0.92	0.9651	-
STS-HGCN-AL [36]	-	0.938	0.955	-
Proposed	97.1696	0.9908	0.9639	0.9661

criticized for their "black-box" nature, the suggested method leverages graph-based representations to provide insights into brain connectivity patterns. It offers a more interpretable framework for predicting the pre-ictal brain activity. This explainability is crucial for medical applications, as it enables clinicians to trust and validate the model's predictions. BNT's performance was also benchmarked on an Nvidia Jetson platform. This is very important for real-time seizure prediction, as it allows for continuous monitoring of patients using wearable EEG devices. Processing data efficiently on edge devices also reduces delay. This can allow for early interventions.

Despite the promising results, this study has some limitations. First, BNT was evaluated on a single dataset (CHB-MIT). Subsequent research could validate the model across more heterogeneous datasets to assess generalizability. Second, while our BNT-based approach provides explainability through graph-based representations, further research is needed to interpret the specific brain connectivity patterns associated with pre-ictal states. Additionally, the computational efficiency of BNT was assessed only on an Nvidia Jetson Xavier NX. Its performance on other edge devices is yet to be explored. Future work could investigate the model's adaptability to other edge platforms.

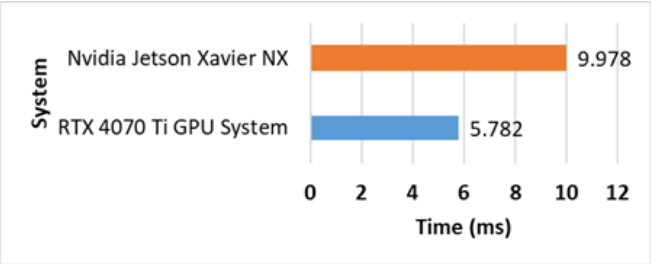

Fig. 2. Inference time analysis of BNT on GPU system and edge device.

VI. CONCLUSION

This study demonstrates the effectiveness of our edge-enabled geometric deep learning based framework for the prei-ctal state prediction of epileptic seizures using EEG data. The proposed BNT-based methodology achieves high accuracy in comparison to existing techniques, making it suitable for medical applications. Benchmarking on an edge device confirms its feasibility for real-time use. Although the results are promising, in future research, we will test BNT on diverse datasets and optimize it for different edge platforms. Additionally, further analysis of brain connectivity patterns identified by BNT could provide deeper insight into seizure mechanisms. This work contributes to advance the prediction of pre-ictal seizures and improve patient care through early intervention, paving the way for more accessible and reliable wearable monitoring systems.

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